

Harsh Dave

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Education

Texas A&M University

M.S. Computer Science

College Station, TX

Expected 2028

Texas A&M University

B.S. Computer Science & B.S. Statistics, Summa Cum Laude

College Station, TX

May 2026 | GPA: 3.92/4.00

Research Interests

Computational genomics of *Mycobacterium tuberculosis*; molecular evolution and positive selection; tuberculosis–diabetes comorbidity; drug resistance.

Research Experience

Graduate Assistant Research (Computational Genomics)

June 2026 – Present

Ioerger Lab, Texas A&M University (Advisor: Thomas R. Ioerger)

College Station, TX

- Investigating evidence of positive selection in *M. tuberculosis* genomes from diabetes and non-diabetes patient cohorts, analyzing approximately **4,000 genes** to identify candidate genomic differences.
- Built reproducible **Python** and **Slurm** pipelines for cohort QC, FASTA processing, Bayesian dN/dS estimation with **GenomeMap**, and posterior comparisons across large-scale TAMU HPRC runs.
- Validated selection signals with pN/pS enrichment, chi-square testing, Benjamini-Hochberg and local-FDR correction, drug-resistance positive controls, and independent **PAML/codeml** site-model analyses.

Undergraduate Researcher

January 2025 – May 2025

UrbanResilience.AI Lab, Texas A&M University

College Station, TX

- Built **Python** ingestion and preprocessing pipelines integrating air-quality sensor APIs and web-scraped wildfire data across Los Angeles sources for downstream emergency-response analysis.

Manuscripts in Preparation

Dave, H., Shatby, A., and Ioerger, T. Differential Bayesian analysis of genomic sequences identifies *M. tuberculosis* genes under positive selection in TB patients with diabetes.

Teaching Experience

Teaching Assistant, CS 111

Spring 2026

Texas A&M University

College Station, TX

- Supported students in **Java** and object-oriented programming through labs, debugging help, and weekly assignment review.
- Reviewed weekly submissions with consistent feedback and explained bugs through questions and small code examples.

Additional Experience

Software Engineering Capstone – Amazon-Sponsored

August 2025 – December 2025

Texas A&M University

College Station, TX

- Built a breach-intelligence analytics platform in a **3-person team**, implementing **Scrapy** and **Tor** ingestion plus a **LangChain** and **Gemini 2.5 Flash Lite** pipeline for entity extraction and threat classification.
- Designed storage and retrieval across **AWS S3**, **MongoDB**, **Elasticsearch**, **Redis**, and **Neptune**; delivered Streamlit and Kibana views for querying unstructured breach data by sector and geography.

Cloud Team Research Member

Early undergraduate

SpaceCraft VR

College Station, TX

- Helped an eight-person research team automate cloud infrastructure for simulation work.
- Contributed to a **React** and **TypeScript** sandbox and hardened access flows with **Auth0**.

AI Engineering Intern

May 2024 – August 2024

Videomagic

Remote

- Developed and evaluated **PyTorch** and **Hugging Face Transformers** computer-vision workflows for AI-generated video detection, analyzing frame-level outputs and model failure modes.
- Built **Feathers.js**, **MySQL**, and **Knex.js** APIs for model inference and data management, secured with **Auth0** and **JWT RBAC**.

Technical Skills

Data Science & Statistics: Python, R, SQL, pandas, NumPy, SciPy, scikit-learn, Bayesian inference, MCMC, hypothesis testing, FDR correction, Matplotlib

Bioinformatics & HPC: GenomeMap, PAML/codeml, dN/dS, pN/pS, FASTA pipelines, Slurm, Linux, TAMU HPRC

Machine Learning & Data Systems: PyTorch, Hugging Face Transformers, LangChain, AWS, MongoDB, Elasticsearch, Redis, Git, GitHub Actions

Programming: Java, C++, TypeScript, JavaScript, React, FastAPI, Docker, PostgreSQL

Languages: English, Hindi, Gujarati

Software

MtbScope

harsh.bet/genes

Search and comparison for the H37Rv genome

- Search across **4,018** H37Rv protein-coding genes and compare several side by side, with annotations, locations, operons, and links back to the sources.
- Ranking treats missing measurements as missing rather than as zero.

Research

harsh.bet/research

Paper reading

- Import open-access papers by DOI, PMID, or PMCID and keep text, figures, tables, equations, and references with the paper.
- Search a saved library, and turn notes into flashcards, quizzes, blanks, and matching exercises on the device.

Quizlet

harsh.bet/quizlet

Exam-style study

- Flashcards, quiz, fill-in-the-blank, and matching. Decks stay on the device unless sync is turned on.

Simplfy

harsh.bet/simplfy

Tutor-style lessons

- Each lesson starts with an analogy and a worked example, then practice, then an explanation in the student's own words.

Radar

harsh.bet/radar

Papers, campus events, and paid studies

- Combines listings from multiple sources, removes duplicates, and keeps a link back to each source.
- Ranks paid studies by guaranteed hourly pay and lists unknown rates separately.

PickLedger

harsh.bet/pickledger

Daily sports picks and source records

- Built and deployed an automated multi-sport prediction and evaluation platform spanning **8 team-model** and **3 player-prop** pipelines with versioned JSON contracts and a Vite/TypeScript frontend.
- Designed immutable pregame ledgers and probability-calibration workflows evaluated with **Brier score**, **log loss**, **ECE**, and **market baselines**; automated grading, validation, and GitHub Pages deployment.

Slate, Daymark, and Notes

Tasks, habits, notes

Private, with optional sync

- **Slate** keeps lists, due dates, and priorities. **Daymark** tracks habits by count, time, or distance, with streaks and reviews.
- **Notes** uses folders, labels, search, and optional rich text. Deleted notes go to trash before permanent deletion.

Fare and Recipes

Meals

Nutrition tracking and a vegetarian meal library

- **Fare** tracks calories and macros with saved foods, barcode search, and meal history. Past entries keep the nutrition values recorded at the time.
- **Recipes** is a searchable library of substantial vegetarian lunches and dinners, with macros, steps, and grocery notes.

Gym, ShotLab, Timer, and Degree

Training and planning

On-device training, form, focus, and degree planning

- **Gym** stores reusable programs, sets, history, and personal records that stay intact when a program changes.
- **ShotLab** estimates basketball shooting form from video on the device. **Timer** is a focus timer with scenes drawn in the browser.
- **Degree** checks catalog rules, credit totals, and research-hour limits as courses move between terms.

Competitions and Course Projects

Alpha, Tidal Hackathon

1st place

React, TypeScript, multi-model tools

- Built a math-learning app with interactive explanations, visualizations, notes, and PDF uploads. Input validation reduced tool-call failures from **25% to 6%**.
- Normalized outputs from Gemini, OpenRouter, and Claude into consistent component states for the demo.

TAMU Datathon

2nd of 50 teams

Web scraping and AI tools

2024

- Used web scraping and AI tools to investigate clues during the competition.
- Placed **2nd of 50 teams** in a prompt-engineering challenge.

Sign Sense

Hackathon

YOLOv5, SageMaker, Svelte, FastAPI

- A sign-language practice app. The team trained a YOLOv5 model on SageMaker and served recognition through FastAPI to a Svelte interface.
- Connected the model to a camera-based practice flow so a learner gets feedback during a lesson.

ProfFinder, HowdyHack

24-hour build

SQL, JavaScript

- Faculty and course search over a database assembled from Texas A&M data, with schedule cards and section search.
- Assembled the course database during the hackathon and used it for class, section, and schedule search.

Point of Sale System

Scrum lead

React, PostgreSQL, AWS, OAuth2

- Led delivery of a point-of-sale platform covering ordering, inventory, analytics, authentication, and accessible customer flows.
- Integrated **OAuth2** and kept customer flows aligned with **WCAG 2.1**.

Coursework

CSCE 671, Computer-Human Interaction

Fall 2026

Texas A&M University

Graduate

- How people use software, how interfaces are designed, and how to evaluate whether a design works.
- Topics include accessibility, usability evaluation, and reading HCI research.

CSCE 627, Theory of Computability

Fall 2026

Texas A&M University

Graduate

- Automata, Turing machines, decidability, and reductions.
- Topics include finite automata, formal languages, and computational complexity.